



Identification and inspection of the vacancy site in Li doped BPO_4 ceramic electrolyte by NMR

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Abstract

A study of the properties of the high temperature ceramic electrolyte $\text{Li}_x\text{B}_{1-x/3}\text{PO}_4$ (lithium boron phosphate) is reported. XRD and NMR are used to investigate changes of the material as a function of heat treatment. It was found that after synthesis at 450 °C the material contains a phase of $\text{Li}_4\text{P}_2\text{O}_7$ in addition to the BPO_4 phase. This second phase is removed by heat treatment at temperatures higher than 600 °C. Boron vacancies are present, REDOR and CPMAS techniques are used to investigate this defect site and show that for the heat treated material Li ions are present at the vacancy site.

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1. Introduction

Li ion rechargeable batteries are class leaders in areas such as cell voltage and energy density, yet they typically rely on liquid or polymer based electrolytes which have an upper temperature limit well below 100 °C. Thus a high temperature electrolyte is required to expand the operating range of these batteries.

The structure and properties of undoped boron phosphate (BPO_4) have been studied since the late 1930's and are well known. In the quest for high performance, inexpensive solid Li ion conducting electrolytes, Kelder et al. [1] demonstrated the fea-

sibility of doping the high-cristobalite boron phosphate crystal with Li ions. This new Li doped boron phosphate ($\text{Li}_x\text{B}_{1-x/3}\text{PO}_4$) could then be incorporated in a battery and submitted to dynamic compaction to improve contact at the interface, leading to a cell with room temperature conductivity of the order of 10^{-4} S/cm [2]. This is comparable with its polymer based counterparts but with the advantage of high temperature capabilities.

The properties of the Li doped boron phosphate have been extensively studied by diffraction and other techniques [3–5]. Two possible defect models have been derived showing the existence of a boron vacancy balanced by interstitial Li ions, shown here using the Kröger–Vink notation



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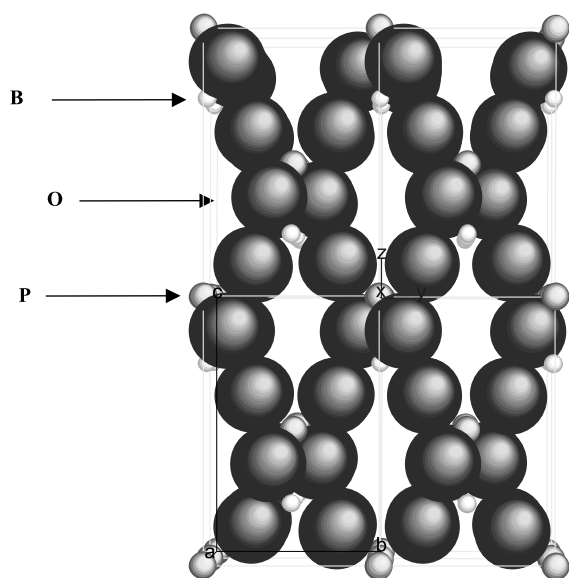


Fig. 1. High cristobalite structure of the undoped boron phosphate. The scale of the oxygen atoms has been reduced slightly to clearly indicate the position of the three-dimensional conduction channels within the structure.

In previous work on these materials it has been assumed that Li ion conductivity takes place via the three-dimensional channels in the structure of BPO_4 which can be seen in Fig. 1. In this work, we study the effects of heat treatment on a series of samples with doping level $x = 0.2$, referred to as 20 mol% hereafter. Samples were synthesized at 450 °C and subsequently heated to 600, 800 and 1000 °C for a controlled period. X-ray diffraction (XRD) is used as a tool to determine the existence of a single crystalline phase. Both the vacancy site and its immediate environment have been investigated using single pulse, magic angle spinning (MAS), cross polarization (CP) and double resonance NMR techniques.

2. Experimental

2.1. Sample preparation

The synthesis route for $\text{Li}_x\text{B}_{1-x/3}\text{PO}_4$ via the phosphorous pentoxide route is described in detail in [1]. Samples were made with $x = 0.2$ and were

initially synthesized at 450 °C. Subsequent heating was carried out for 10 h at 600, 800 and 1000 °C following the same heating regime as that used in synthesis, each sample to be heat-treated was taken from the originally synthesized powder. Due to the hygroscopic nature of BPO_4 , the samples were stored in a dessicator at all times and exposure to the atmosphere was kept to a minimum.

2.2. Single pulse NMR

The ^{31}P MAS-NMR, ^7Li NMR and ^{11}B single pulse spectra were obtained using a Chemagnetics CMX-300 spectrometer, giving resonance frequencies of 121.51, 116.68, and 96.37 MHz, respectively. A 90° pulse length of 2 μs (rf field strength of 125 kHz) was achieved for both the ^{31}P and ^7Li nuclei, and 3.2 μs for ^{11}B (rf field strength of 78 kHz), using a 4 mm HX double resonance MAS probe. Spinning speed of 13 kHz was employed for the above MAS experiments. VPI-5 was used as an external reference for the ^{31}P NMR with the central peak assigned to -27.5 ppm. LiCl (1 M) solution was used for the ^7Li reference at 0 ppm and H_3BO_3 provided the reference for the ^{11}B , also at 0 ppm.

2.3. REDOR

$^{31}\text{P}\{^6\text{Li}\}$ REDOR (rotational echo double resonance) experiments were carried out on a Chemagnetics CMX-600 spectrometer using a Varian T3 triple resonance probe – this gave a ^{31}P resonance frequency of 242.92 MHz and a ^6Li frequency of 88.32 MHz. In this case the ^{31}P 90° pulse length was 2.2 μs and the ^6Li pulse length was 2.9 μs (rf field strength of 114 and 86 kHz, respectively). REDOR experiments were carried out at a spinning speed of 13.85 kHz.

In the REDOR experiment [6,7] a rotor synchronous echo sequence is applied to the observed nucleus (I spin, ^{31}P in this case) while two 180° pulses per rotor period are applied to another nucleus (S spin, in this case ^6Li). If the S spin is in close proximity to the I spins then the observed I spin echo intensity is reduced during the echo. This occurs because the 180° pulses on the S spin change the sign of the heteronuclear dipolar coupling (with

the I spin). The result of this is a disturbance in the refocusing process which would normally occur under the combined action of MAS and an echo sequence. The relative amount of signal lost increases with longer echo times and shorter I–S internuclear distances (i.e., stronger heteronuclear dipolar coupling). In this Letter, the $^{31}\text{P}\{^6\text{Li}\}$ REDOR is used as a qualitative tool rather than to obtain an exact distance measurement.

2.4. XRD

XRD measurements were made on a Bruker AXS X-ray diffractometer with a Cu-K α radiation source. The range $90^\circ < 2\theta < 10^\circ$ was inspected in step sizes of 0.01° and step time of 1.3 s. Particle sizes were derived from the 75.5° (3 1 2) reflection using the Scherrer equation [8].

3. Results and discussion

The XRD spectra taken for the 20 mol% sample after synthesis at 450°C , and further heat treatment at 600 , 800 and 1000°C , are shown in Fig. 2.

The spectra show a diffraction pattern that is typical for the boron phosphate cristobalite structure. However, small reflections at 20.5° and 28° in the spectra at 450 and 600°C can be discerned which are indicative of the presence of a second phase material – in this case $\text{Li}_4\text{P}_2\text{O}_7$. After treatment at higher temperatures this extra phase disappears and the cristobalite pattern remains. LiBPO_4 is used as a solid electrolyte where the conducting Li ions are expected to be in the BPO_4 structure, traveling through the channels in the crystal. The extra phase containing Li ions is therefore of some concern as conduction could take place through the extra phase instead of in the bulk material. During the course of the investigation it was noted that brief exposure to moisture in the atmosphere led to marked changes in the properties of the material. After just a few minutes of exposure the XRD spectra for the 450°C material displayed further peaks due to evolution of $\text{B}(\text{OH})_3$. More importantly, the ^7Li NMR shows that a brief exposure to the atmosphere almost halves the room temperature Li linewidth from 5.5 to 3 kHz. This strongly indicates a great increase in Li ion mobility for *all* of the Li ions. In contrast,

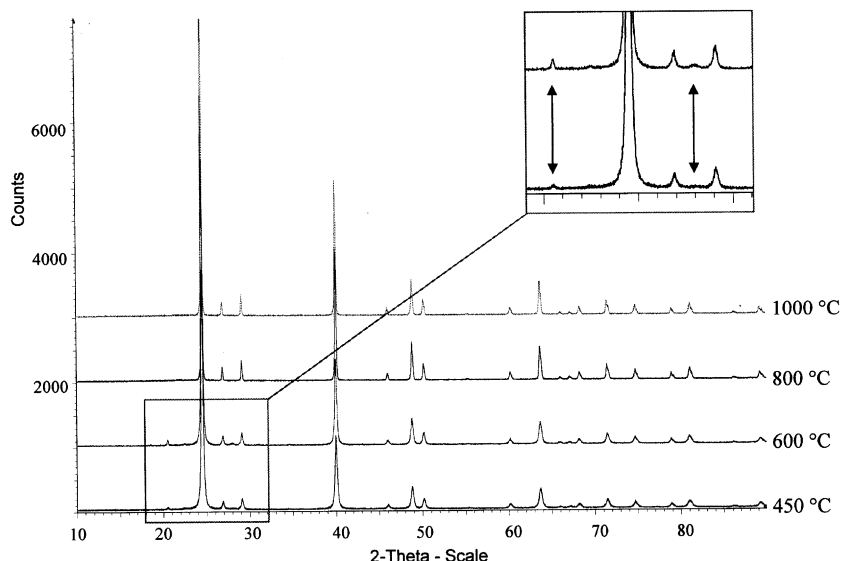


Fig. 2. XRD spectra for the 20 mol% Li boron phosphate. The two lower spectra are for the samples synthesized at 450 and 600°C and clearly shows the impurity peaks at $2\theta = 20.5^\circ$ and 28° due to the presence of an $\text{Li}_4\text{P}_2\text{O}_7$ phase. Above this are shown spectra for the samples heat treated to 800 and 1000°C , respectively.

for the material treated at temperatures higher than 600 °C the Li linewidth is unaffected by exposure to the atmosphere. It is clear that for the material synthesized at lower temperatures, the solid-state reaction is incomplete and that the Li ions can be found at the surface of the material. This indicates the possibility that the conduction in these impure materials is a surface effect aided by the impurities in the material – Li ions are conducting through impurity layers at the surface of the BPO_4 material, and not through the channels in the cristobalite structure as was previously thought. For the high-temperature material however, it seems that all the Li ions are now within the BPO_4 structure.

Fig. 3 shows the ^{31}P NMR spectra with MAS at 13 kHz for the undoped sample treated to 1000 °C along with the 20% doped sample treated to both 450 and 1000 °C. The spectrum for the 20% sample treated to 450 °C shows a main peak at –30 ppm and some very small impurity peaks for samples treated at low temperature. This peak at –30 ppm is also observed in the ^{31}P spectrum of the und-

oped boron phosphate material and so represents the bulk BPO_4 site. At 1000 °C a very interesting feature develops at around –10 ppm. A broad peak of ‘amorphous’ nature appears alongside the main peak. This peak is of particular interest since its intensity was found to vary proportionally to the Li doping level (Fig. 3, inset), indicating that it is directly related to the Li ions incorporated within the structure after heat treatment. There is no sign of any extra peak in the ^{11}B NMR spectrum which suggests that the interaction which is causing the peak is local to the ^{31}P nucleus.

The defect model derived for this material (Eq. (1)) indicates boron vacancies which are charge balanced by the Li ions in the structure. However, since the XRD data and the narrowing of the lithium linewidth in the presence of water indicate that all Li ions exist in a separate phase at 450 °C, this cannot be the case. It is more likely that the vacancy is charge balanced by protons, introduced into the structure during synthesis. In order to rule out the possibility that the broad feature in the spectrum of the 1000 °C sample is related to

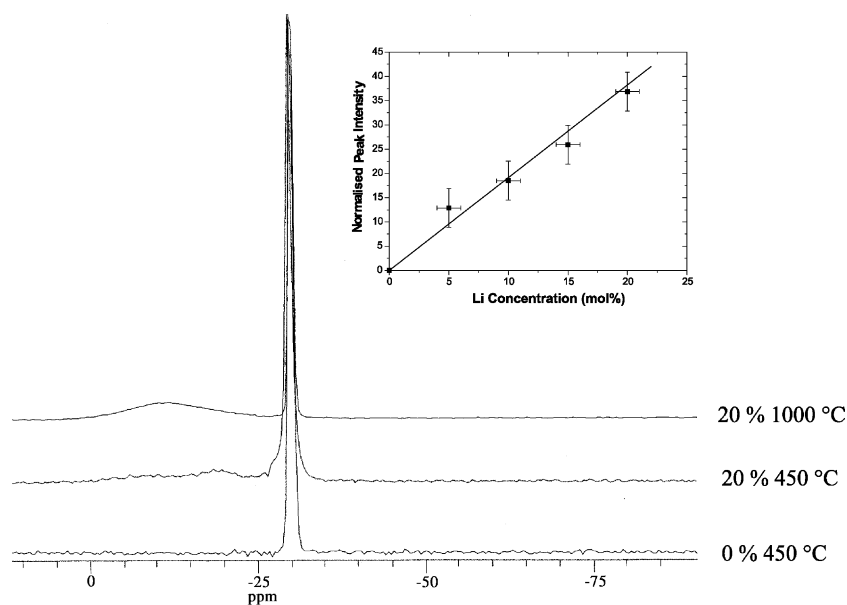


Fig. 3. ^{31}P MAS NMR spectra at 13 kHz for the undoped material at 1000 °C and the 20 mol% material at 450 (particle size 30 nm) and 1000 °C (particle size 60 nm). The broad feature shown in the 1000 °C spectrum indicates a region of high chemical shift anisotropy. Inlaid is the development of this peaks intensity with increasing Li concentration. Note: intensities do not exactly match theoretical values due to the extremely long T_1 (>1 h) of the bulk ^{31}P site.

protons CP was used in an attempt to transfer magnetization from the ^1H nuclei to nearby ^{31}P and ^{11}B nuclei. Fig. 4 shows the results of these experiments for the material synthesized at 450 °C and that heat treated to 1000 °C. For the ^{11}B nucleus it was possible to transfer magnetization in the 450 °C material, however, only a noise signal

was recovered at 1000 °C. The same was observed for the ^{31}P nucleus, but interestingly the signal recovered at 450 °C shows both the broad peak and the main BPO_4 peak. This suggests that protons are occupying the same site in the 450 °C sample as is occupied by Li in the 1000 °C sample. The failure to transfer magnetization in the 1000 °C

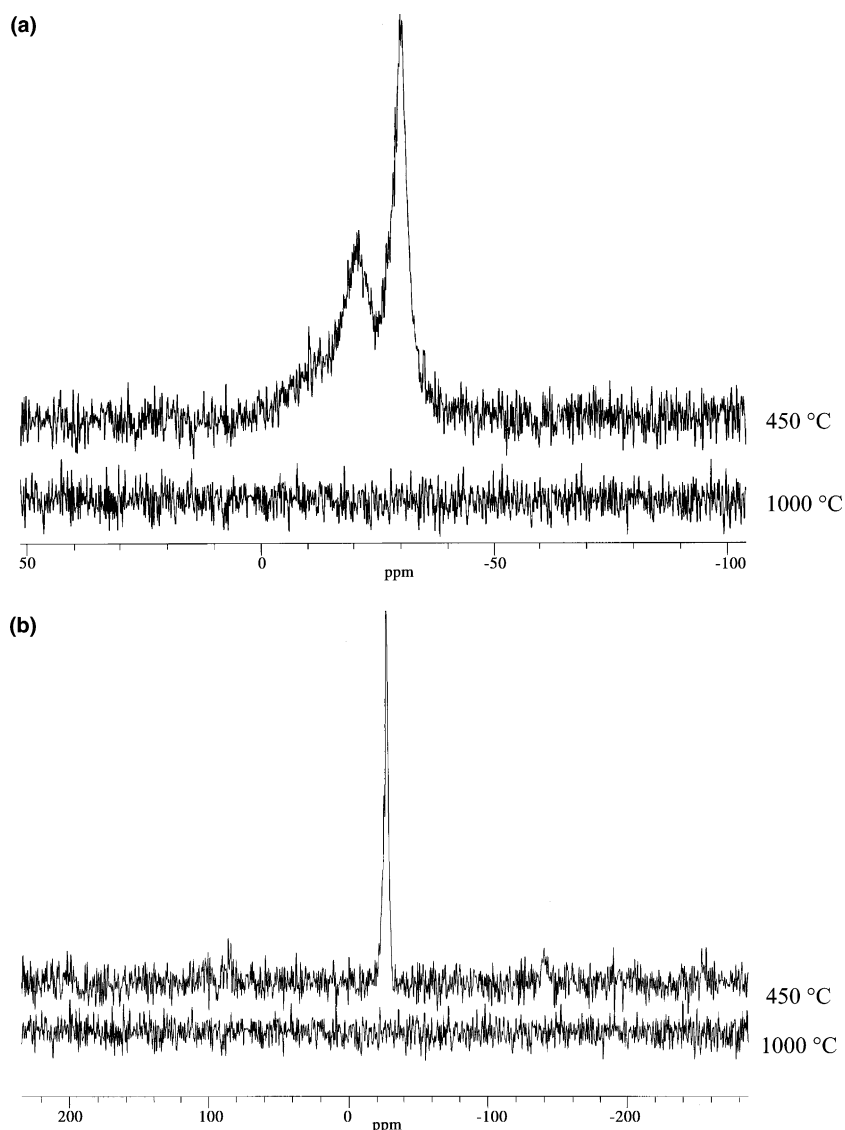


Fig. 4. (a) ^1H - ^{31}P CPMAS for the sample synthesized at 450 °C and that heat treated to 1000 °C. Magnetization was successfully transferred to both P sites for the lower temperature sample yet there was no evidence of protons in the heat treated material. (b) ^1H - ^{11}B CPMAS for samples as in (a).

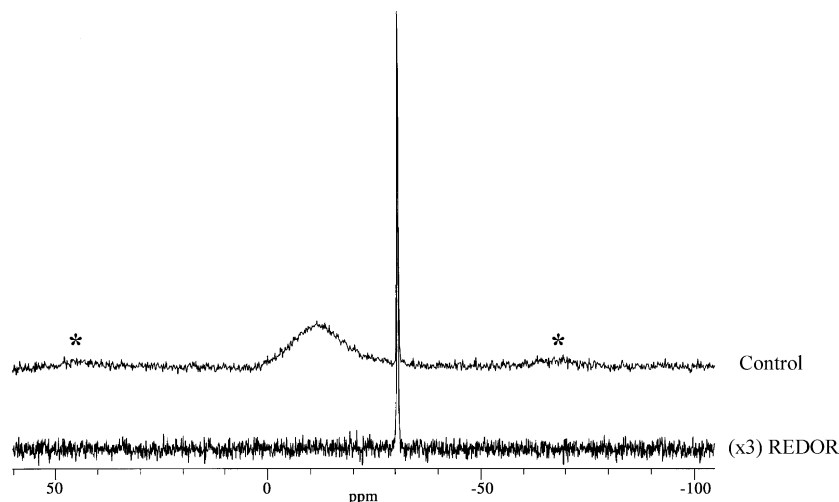


Fig. 5. REDOR result for 20 mol% sample spinning at 13.85 kHz. Shown is the result of the simple ^{31}P (control) echo with the Li rf channel inactive, and the result of the REDOR experiment (magnified $3\times$ for clarity) showing that the broad peak is the result of ^6Li nuclei in the vicinity of ^{31}P .

sample is proof that the broad feature observed in its ^{31}P MAS spectrum is not related to protons. It also demonstrates that protons which were present after initial synthesis are eradicated from the structure following heat treatment.

In order to investigate the possibility that this broad feature in the ^{31}P spectrum is related to Li ions in the structure, a $^{31}\text{P}\{^6\text{Li}\}$ REDOR experiment was employed. Samples were prepared using ^6Li (94%) to overcome the experimental difficulties inherent in double resonance where nuclei have similar resonance frequencies such as ^{31}P and ^7Li . These samples exhibited the same features in the XRD and NMR as their equivalent ^7Li synthesized material. The spectra shown in Fig. 5 show the normal rotor-synchronized control echo along with the spectrum obtained using the REDOR sequence. In comparing the spectra it is clear that the broad peak at -10 ppm must represent phosphorus atoms in close proximity to lithium since this resonance is completely destroyed by the REDOR sequence. Therefore, the linewidth of the ^7Li static spectrum (not shown) originates from lithium ions at the defect site. The rigid lattice value for the ^7Li linewidth of 5.5 kHz can only be obtained if clusters of Li ions are considered [3].

4. Conclusions

Using both XRD and NMR techniques it has been possible to detect a vacancy site within the boron phosphate matrix of Li doped BPO_4 . Furthermore Li ions are shown to be present at this vacancy site, which is expected if the material is to be charge balanced. However, this vacancy site is only occupied by Li ions after further heat treatment of the material beyond its synthesis temperature of 450°C . Before heat treatment above 600°C it is shown that the Li ions are present in a separate phase within the material, evidence then suggests that the boron vacancy site must be balanced by protons incorporated within the structure, which are subsequently removed by heat treatment.

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